Experiment M11bis
Torsional Pendulum

Victoria Torres Ochando and Martina Lodos Falcón
Group B4-1

Laboratory session 13/11/2025

Contents

1 Introduction **2**

2 Results and discussion **2**
2.1 Determination of the torsion constant 2
2.2 Determination of the moment of inertia of different bodies 4
2.3 Comparison of experimental and theorical moments of inertia 4
2.4 Experimental verification of the parallel axis theorem (Steiner's theorem) 5

3 Conclusion **7**

4 Appendixes **7**
4.1 General Uncertainties Calculations 7
4.2 Specific Uncertainty Calculations 7
4.2.1 Torsion constant D 7
4.2.2 Determination of the moment of inertia I_z 8
4.2.3 Determination of the moment the theorical moments of inertia I_z 8
4.2.4 Steiner's Theorem Parameters 9
4.2.5 Expanded Uncertainties and Confidence Intervals 9
4.2.6 Input Parameter Uncertainties 9

1 Introduction

This laboratory experiment investigates rotational dynamics using a torsional pendulum system. The primary objectives are to determine the torsion constant of a wire and calculate moments of inertia for various rigid bodies including a rod, disk, sphere, and hollow cylinder. The study also experimentally verifies Steiner's theorem, which describes how moment of inertia changes with parallel axis displacement. By comparing experimental measurements with theoretical predictions and analyzing oscillation periods at different rotation axes, this work validates fundamental principles of rotational mechanics while assessing the precision of experimental measurement techniques.

2 Results and discussion

2.1 Determination of the torsion constant

In this section, the torque required to maintain the yellow disk in a state of equilibrium was measured using a dynamometer. Measurements were taken for various angular displacements by rotating the disk both clockwise and counterclockwise to positions of $\theta = [\frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi]$ radians. The data obtained in the laboratory is organized in Table 1.

Table 1: Values of $\Delta\theta$, τ and their corresponding uncertainties.

$\Delta\theta$ **(rad)**	τ **(N·m)**
1,571 ± 0,050	-0,0400 ± 0,0066
3,142 ± 0,050	-0,0700 ± 0,0066
4,712 ± 0,050	-0,1250 ± 0,0066
6,283 ± 0,050	-0,1550 ± 0,0066
-1,571 ± 0,050	0,0400 ± 0,0066
-3,142 ± 0,050	0,0800 ± 0,0066
-4,712 ± 0,050	0,1200 ± 0,0097
-6,283 ± 0,050	0,1500 ± 0,0043

Based on the experimental measurements, two distinct plots were generated: one corresponding to positive angular displacements and another for negative angular displacements.

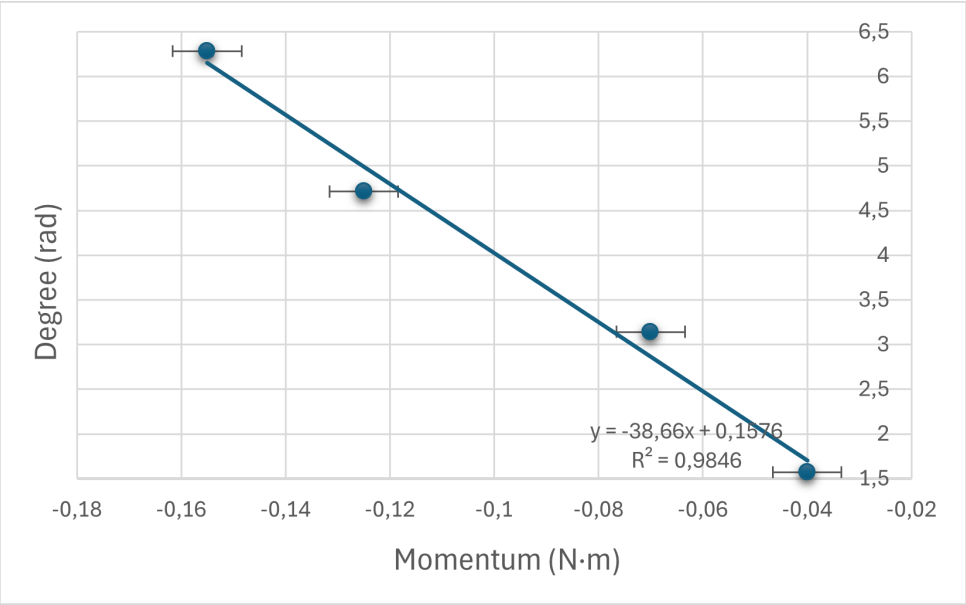


Figure 1: Relationship between angular displacement $\Delta\theta$ and applied torque τ for positive angular increments.

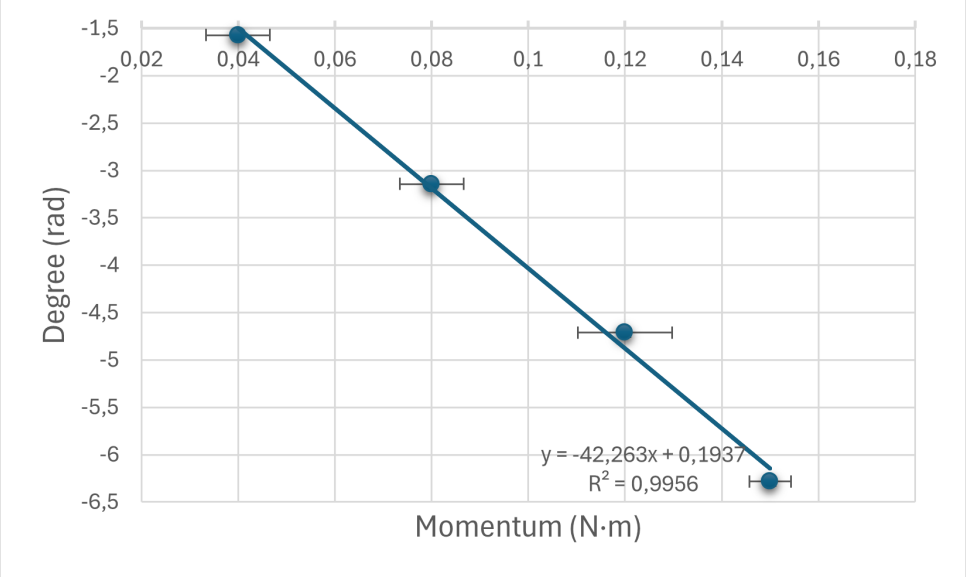


Figure 2: Relationship between angular displacement $\Delta\theta$ and applied torque τ for negative angular increments.

As we can see, the value of the coefficient of determination, R^2, is closer to one in the second graph, indicating a superior linear correlation. Therefore, the parameters derived from this dataset were selected as the most accurate representation of the physical system. The linear regression analysis yields the following relationship:[1]

$$\Delta\theta\,(\tau) = (-42,26 \pm 0,30)\tau + 0,1937, \tag{1}$$

where the slope represents the inverse of the torsion constant, and, consequently,

$$D = 0,02366 \pm 0,00017\ (N \cdot m \cdot rad^{-1}). \tag{2}$$

[1]**The uncertainty of the slope was obtained using a linear regression program**.

2.2 Determination of the moment of inertia of different bodies

In this part of the experiment we measured the period of oscillation for various rigid bodies. The objects tested were a rod, a graduated disc, a sphere and a hollow cylinder. The period was measured following release from a set angle, θ. The results demonstrated that the period is independent of the initial angular displacement and the direction of rotation.

In addition, the physical characteristics of each object were quantified to support the analysis in the following subsection.

Once the data was collected and organized in ([1]), we were able to calculate the moment of inertia for each rigid body. To do so, we applied the relation:

$$I = \frac{T^2 \cdot D}{4\pi^2}. \tag{3}$$

The results, obtained by applying the arithmetic mean to the set of I_z values for each angular displacement, are presented in Table 2.

Table 2: Moment of inertia calculated for each rigid body with its corresponding uncertainties.

Rigid body	**Rod**	**Graduated disk**	**Solid sphere**	**Hollow cylinder**
$I_z \quad (kg \cdot m^2)$	0,0017 ± 0,0011	0,0042 ± 0,0051	0,0020 ± 0,0017	0,00091 ± 0,00084

2.3 Comparison of experimental and theorical moments of inertia

Once the experimental moments of inertia obtained for the body had been calculated, we obtained its theoretical value using the standard formulas for each geometry and their measured masses and dimensions (Figure 3).

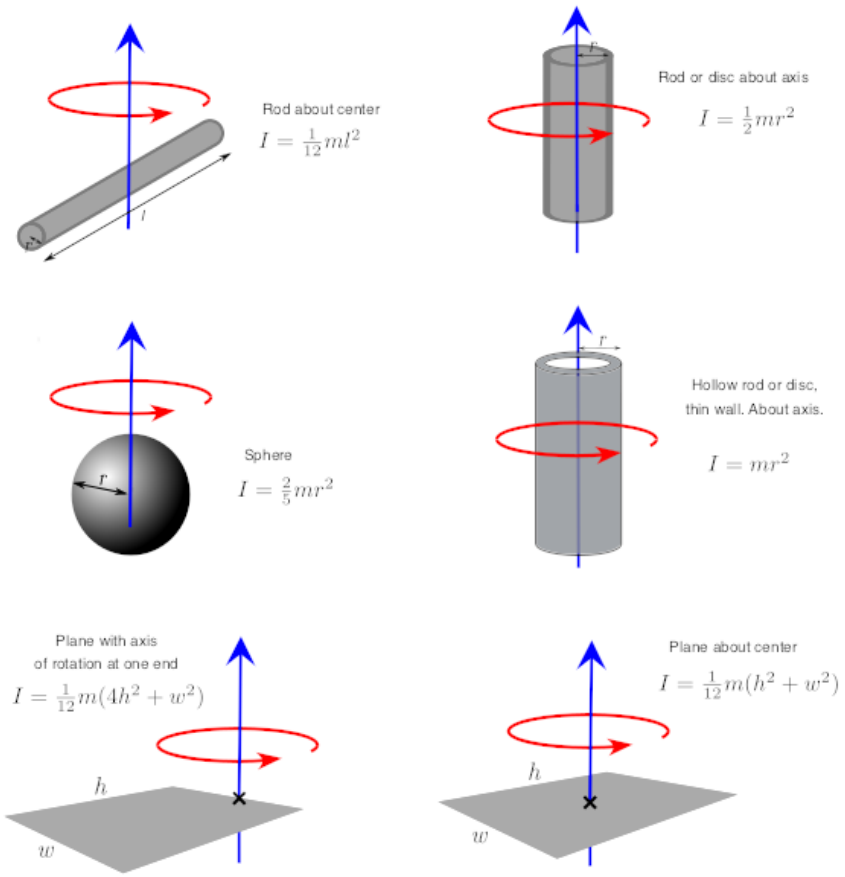


Figure 3: Moments of inertia for different bodies.

The results are summarized in Table 3, which also includes the relative error for each body. The experimental values show good agreement with the theoretical predictions for most bodies, with relative errors below 30.%.

Table 3: Comparison of experimental and theoretical moments of inertia (the uncertainties [2] represents the expanded ones).

	Iz(theor.)$(kg \cdot m^2)$	**Uc(Iz)**$(kg \cdot m^2)$	**Iz(exper.)**$(kg \cdot m^2)$	**Uc(Iz)**$(kg \cdot m^2)$	**Rel. error**(%)
Graduated disk	0,004434	0,000033	0,004	0,010	5,0
Rod	0,001545	0,000026	0,0011	0,0022	27
Solid sphere	0,0015939	0,0000050	0,0020	0,0034	23
Hollow cylinder	0,000961	0,000022	0,0009	0,0017	5,6

In all cases, the theoretical values fall within the 95% confidence intervals of the experimental measurements, confirming the validity of the theoretical models for calculating moments of inertia.

2.4 Experimental verification of the parallel axis theorem (Steiner's theorem)

To verify Steiner's Theorem, we used the graduated disk and measured its oscillation period for five different positions of the axis of rotation, each at a different perpendicular distance a from the disk's center of mass.
For each position, the disk was rotated, and its period was measured with the photogate.
The data collected is presented in Table 4.

Table 4: Relevant results for the described experiment (the complete table is found in the Excel file [1].

Distance		**Angle**		**Squared Distance**		**Squared Period**	
a (m)	**uC(a) (m)**	θ **(rad)**	**uC(**θ**) (rad)**	a^2 **(m^2)**	**uC(**a^2**) (m^2)**	T^2 **(s^2)**	**uC(**T^2**) (s^2)**
0,00000	0,00029	1,571	0,050	0	0	8,0769	0,0037
0,00000	0,00029	3,142	0,050	0	0	7,8755	0,0025
0,00000	0,00029	4,712	0,050	0	0	7,7284	0,0036
0,03000	0,00029	-1,571	0,050	0,000900	0,000017	8,7695	0,0055
0,03000	0,00029	0,785	0,050	0,000900	0,000017	8,8645	0,0026
0,03000	0,00029	-1,047	0,050	0,000900	0,000017	8,8149	0,0038
0,06000	0,00029	0,524	0,050	0,003600	0,000035	10,9848	0,0029
0,06000	0,00029	0,785	0,050	0,003600	0,000035	10,9340	0,0061
0,06000	0,00029	-0,349	0,050	0,003600	0,000035	10,9826	0,0043
0,09000	0,00029	0,1745	0,050	0,008100	0,000052	14,8379	0,0049
0,09000	0,00029	0,524	0,050	0,008100	0,000052	14,7686	0,013
0,09000	0,00029	0,262	0,050	0,008100	0,000052	14,8276	0,0034
0,12000	0,00029	0,785	0,050	0,014400	0,000069	17,4139	0,019
0,12000	0,00029	0,524	0,050	0,014400	0,000069	17,3195	0,0077
0,12000	0,00029	-0,262	0,050	0,014400	0,000069	17,1589	0,0077

A linear regression fit of the form $T^2 = slope \cdot a^2 + intercept$ was performed with this pair of data. The following graph represents the results of that fit.

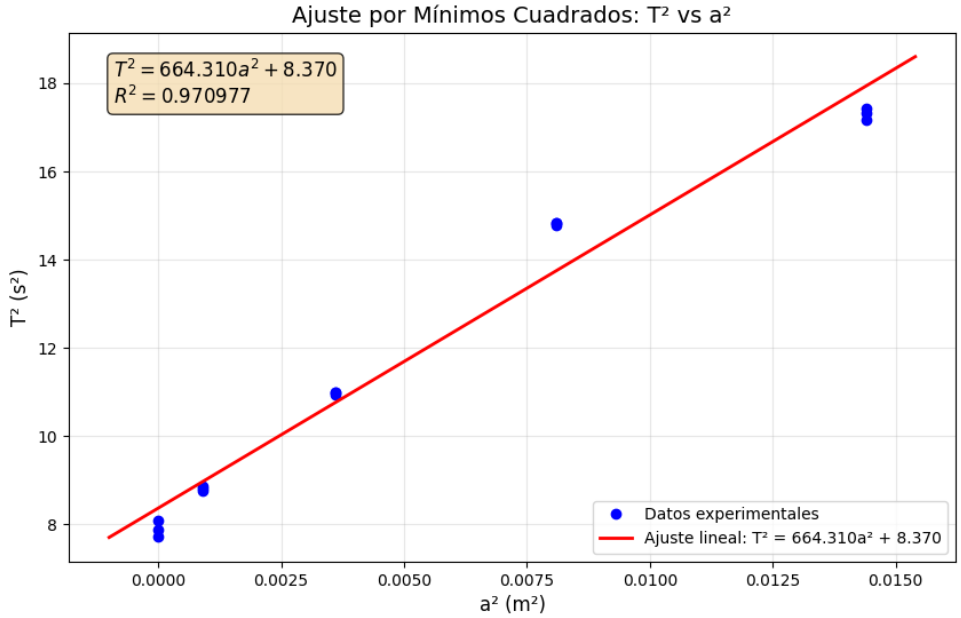


Figure 4: Linear regression fit for T^2 and a^2 data.

As we can see, the slope takes a value of $6,6 \cdot 10 \pm 32 s^2/m^2$, while the intercept gets $8.4 s^2$. The high value of $R^2 = 0,97$ confirms a strong linear relationship, as predicted by Steiner's Theorem.

On the other hand, according to the theoretical framework [3], the relationship between the squared period T^2 and the squared distance a^2 is given by:

$$T^2 = \frac{4\pi^2}{D}(I_{cm} + ma^2) \tag{4}$$

which can be expressed as a linear function:

$$T^2 = \frac{4\pi^2 m}{D} \cdot a^2 + \frac{4\pi^2 I_{cm}}{D} \tag{5}$$

where the slope is $\frac{4\pi^2 m}{D}$ and the y-intercept is $\frac{4\pi^2 I_{cm}}{D}$.

From that, we calculated the theoretical slope's and intercept's values so we could be able to compare them with the experimental ones. The results obtained are shown in the next table (table 5).

Table 5: Comparation of theoretical and experimental parameters and their corresponding uncertainties [2]

Parameter	**Theoretical**	**uC (Theor.)**	**Experimental**	**uC (Exp.)**	**Rel. Error (%)**
Slope (s^2/m^2)	656	10	664	69	1,2
Intercept (s^2)	7,4	4,8	8,37	0,52	13

Furthermore, as the table shows, the appropriate confidence intervals for the fitted parameters were determined. As the theoretical values for both the slope and the intercept fall within the 95% confidence intervals of the experimental parameters, we can accept the values obtained:

$$slope \in [595, 733]$$

$$intercept \in [7.85, 8.89]$$

3 Conclusion

In conclusion, this experiment successfully demonstrated the principles of rotational dynamics using a torsional pendulum. The experimental moments of inertia for all tested bodies showed good agreement with their theoretical values, as all theoretical predictions fell within the 95% confidence intervals of the experimental results. However, the experimental values and the theoretical ones are not quite the same due to the limitations of the experiment: the rigid bodies were not ideal, so they do not agree fully with the general equations.

Furthermore, the linear relationship between the squared period and the squared distance in the Steiner's theorem verification, supported by a high coefficient of determination ($R^2 = 0.97$), confirms the validity of Steiner's theorem. Therefore, we can affirm the results clearly validate the theoretical models for calculating moments of inertia.

4 Appendixes

4.1 General Uncertainties Calculations

As established in measurement uncertainty analysis, Type A uncertainty, denoted as $u_A(x)$, quantifies the statistical variation associated with repeated observations and is calculated using the equation:

$$u_A(x) = \frac{\sigma(x)}{\sqrt{N}}, \tag{6}$$

where $\sigma(x)$ represents the standard deviation and N the number of observations. On the other hand, Type B uncertainty, $u_B(x)$, is derived from the estimated accuracy of the measuring instrument and is obtained from:

$$u_B(x) = \frac{\delta_x}{\sqrt{12}}, \tag{7}$$

denotes the instrument's resolution ($\delta_V = 0.01$ V for voltage measurements and $\delta_T = 0.001$ s for the photogate period measurements). The combined standard uncertainty, $u_C(x)$, is subsequently expressed as the root sum of squares of these components:

$$u_C(x) = \sqrt{u_A^2(x) + u_B^2(x)}. \tag{8}$$

For derived quantities obtained through functional relationships $y = f(x_1, x_2, \ldots, x_n)$, the uncertainty was propagated using:

$$u_C(y) = \sqrt{\sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}\right)^2 u_C(x_i)^2}. \tag{9}$$

These formulas were extracted from the document [2].

4.2 Specific Uncertainty Calculations

4.2.1 Torsion constant D

The torsion constant is a derived quantity obtained as follows.

$$D = \frac{1}{a}, \tag{10}$$

where a is the slope represented in Figure 2. Therefore, its uncertainty is

$$u_C(D) = \sqrt{\left(\frac{-1}{a^2}\right)^2 \cdot (u_C(a))^2}. \tag{11}$$

4.2.2 Determination of the moment of inertia I_z

Firstly, we need to calculate the errors of the period T taken for each $\Delta\theta$. The angle displacement was measured only once, whereas the period was measured thrice. Then,

$$u_C(\Delta\theta) = u_B(\Delta\theta) = \frac{\pi/18}{\sqrt{12}} = 0,050\ rad, \tag{12}$$

$$u_C(T) = \sqrt{u_A(T)^2 + u_B(T)^2} = \sqrt{\left(\frac{\sigma(T)}{\sqrt{3}}\right)^2 + \left(\frac{0,001}{\sqrt{12}}\right)^2}\ \ (s). \tag{13}$$

We also need to calculate the error of T^2, as it will be used later to determine the moment of inertia of each rigid body.

$$u_C(T^2) = \sqrt{(2T)^2 \cdot u_C(T)^2}\ \ (s^2). \tag{14}$$

Finally, the error for the moment of inertia for each angle displacement $\Delta\theta$ is given by

$$u_C(I_z) = \sqrt{\left(\frac{D}{4\pi^2}\right)^2 \cdot u_C(T^2)^2 + \left(\frac{T^2}{4\pi^2}\right)^2 \cdot u_C(D)^2}\ \ (kg \cdot m^2), \tag{15}$$

and the uncertainty of average value of the moments of inertia I_z for each $\Delta\theta$, calculated by applying the arithmetic mean, is:

$$u_C(\overline{I_z}) = \frac{1}{3}\sqrt{u_C(I_z(\Delta\theta_1))^2 + u_C(I_z(\Delta\theta_2))^2 u_C(I_z(\Delta\theta_3))^2}\ \ (kg \cdot m^2). \tag{16}$$

4.2.3 Determination of the moment the theorical moments of inertia I_z

For theoretical moments of inertia, the uncertainty was propagated from the measured parameters:

- **Disk/Sphere:** $I = \frac{1}{2}mR^2(kg \cdot m^2)$

$$u_C(I) = \sqrt{\left(\frac{1}{2}R^2\right)^2 u_C(m)^2 + (mR)^2\, u_C(R)^2}(kg \cdot m^2)$$

- **Rod:** $I = \frac{1}{12}mL^2(kg \cdot m^2)$

$$u_C(I) = \sqrt{\left(\frac{1}{12}L^2\right)^2 u_C(m)^2 + \left(\frac{1}{6}mL\right)^2 u_C(L)^2}(kg \cdot m^2)$$

- **Hollow cylinder:** $I = mR^2(kg \cdot m^2)$

$$u_C(I) = \sqrt{(R^2)^2\, u_C(m)^2 + (2mR)^2\, u_C(R)^2}(kg \cdot m^2)$$

For experimental moments of inertia:

$$I_{\text{exp}} = \frac{T^2 D}{4\pi^2}(kg \cdot m^2)$$

$$u_C(I_{\text{exp}}) = \sqrt{\left(\frac{D}{4\pi^2}\right)^2 u_C(T^2)^2 + \left(\frac{T^2}{4\pi^2}\right)^2 u_C(D)^2}(kg \cdot m^2)$$

4.2.4 Steiner's Theorem Parameters

For the linear fit $T^2 = slope \cdot a^2 + intercept$:

- **Theoretical slope:** $slope_{\text{theor}} = \dfrac{4\pi^2 m}{D}$

$$u_C(slope_{theor}) = \frac{4\pi^2}{D} \cdot \sqrt{u_C(m)^2 + \left(\frac{m \cdot u_C(D)}{D}\right)^2}$$

- **Theoretical intercept:** $intercept_{\text{theor}} = \dfrac{2\pi^2 m R^2}{D}$

$$u_C(intercept_{\text{theor}}) = \frac{2\pi^2}{D} \cdot \sqrt{(R^2 u_C(m))^2 + (2mRu_C(R))^2 + \left(\frac{mR^2 u_C(D)}{D}\right)^2}$$

4.2.5 Expanded Uncertainties and Confidence Intervals

The expanded uncertainties at 95% confidence level were calculated using:

$$U_{95\%} = t_\nu \cdot u_C(x)$$

with the following t-Student values:

- **Experimental averages (3 measurements):** $\nu = 2$, $t = 4.303$
- **Linear fit parameters (15 points):** $\nu = 13$, $t = 2.160$
- **Theoretical values that depend only on m and R (one measure):** $\nu \to \infty$, $t = 1.960$

4.2.6 Input Parameter Uncertainties

The following standard uncertainties were used in all calculations:

- Mass: $u_C(m) = 0.0001$ kg (from balance specification)
- Radius: $u_C(R) = 0.0005$ m (from caliper resolution)
- Length: $u_C(L) = 0.001$ m (from ruler resolution)
- Torsion constant: $u_C(D) = 0.0071$ N·m/rad (from R1 linear fit)
- Period: $u_C(T)$ from Equation (8) with $\delta_T = 0.001$ s

References

[1] Martina Lodos Falcón and Victoria Torres Ochando. "Hoja de cálculo". 2023. URL: https://docs.google.com/spreadsheets/d/113ASqPethzgUg6E47rKGfQF7jbUZdStibQoBnXuLv8c/edit?usp=sharing.

[2] Department of Applied Physics; University of Granada. *Notes for the Calculation of Uncertainties in Laboratory Sessions*. PRADO, 2025.

[3] Miguel Ángel Rodríguez Valverde; Artur Schmitt; José Callejas Fernández; Miguel Cabrerizo Vílchez. *MOVIMIENTO DESDE SISTEMAS EN ROTACIÓN*. Department of Applied Physics, University of Granada, 2023.